

Ques 7) Show that the language pal of palindromes is not regular.

$$\rightarrow L = \{w \mid w = w^R\}$$

using Pumping Lemma

- Take String : ww^R
- Pumping breaks Symmetry $\rightarrow$ not palindromic
- $\therefore$ Therefore : palindromes is not regular

Ques 8) Find CFN generating the language of even string in $\{a, b\}^*$ which the two middle equal.

- $\rightarrow$
- Language over $\{a, b\}^*$ with even length same middle symbols.

$$S \rightarrow aSa \mid bSb \mid aTa \mid bTb$$

$$T \rightarrow aa \mid bb$$

